

Week 8: Learning – Maximum Likelihood Estimation

Practical Assignment

From this week until the end of the module we look at *learning* from data for PGMs. The theme for this week is maximum likelihood estimation (MLE) of the parameters of a Bayes network (BN) for the case where all the RVs are observed. The assignments consists of two questions:

- The first question estimates the parameters of BN with discrete RVs – the familiar Hamming (7,4) error-correction problem.
- The second question estimates the parameters of BN with continuous RVs – a new problem setup where you have to find the heights of the endpoints of a line from noisy measurements on the line.

1 MLE for the Hamming (7,4) error-correction code

For this question, we revisit the familiar Hamming (7,4) error-correction problem, but we now assume that the following parameters are unknown:

- The probability that any transmitted bit b_i will “flip” to form the received bit r_i is an unknown parameter, θ_e . Note that we use the same parameter for all transmitted/received bits. The probability that any bit will not flip is therefore $1 - \theta_e$.
- The way in which the check bits, $b_4, \dots, b_6$, are generated given the data bits, $b_0, \dots, b_3$, is unknown. For example, the encoding scheme could use even parity or odd parity, but we do not know which. The parameters of the probability distributions, $p(b_4|b_0, b_1, b_2)$, $p(b_5|b_0, b_2, b_3)$ and $p(b_6|b_0, b_1, b_3)$ – i.e., $\theta_{b_4|b_0, b_1, b_2}$, $\theta_{b_5|b_0, b_2, b_3}$ and $\theta_{b_6|b_0, b_1, b_3}$ respectively – are therefore unknown. Note that there are 8 parameters for each table, since e.g. $\theta_{b_4=1|b_0=0, b_1=0, b_2=0} = 1 - \theta_{b_4=0|b_0=0, b_1=0, b_2=0}$. Assume that the parameters of the check bit factors lie in the range $0 \leq \theta \leq 1$, and that the parameters are not shared between factors¹.

The objective of this question is to use maximum likelihood estimation to learn the parameter for of this model. Since we do not need `emdw`'s functionality for this question, you do not have to use C++ for coding (you can e.g. use Python). [To help you check your work, the answer at an intermediate point is given at the end of this question in blue.](#)

- (On paper) Draw the BN (not an MN!) for the model with unknown parameters and for M data points, using plate notation and explicitly showing the parameters. Remember that the variable θ_e is shared among all the factors over the received bits.
- (In code) Choose (actual) values for the parameters $\Theta = (\theta_e, \theta_{b_4|b_0, b_1, b_2}, \theta_{b_5|b_0, b_2, b_3}, \theta_{b_6|b_0, b_1, b_3})$. Now draw M i.i.d. data points from this model. Note that a single data point consist of a combination of values for *all* the RVs in the model; i.e., $\mathbf{d}^{[m]} = (b_0^{[m]}, \dots, b_6^{[m]}, r_0^{[m]}, \dots, r_6^{[m]})$. Do the sampling in stages: first sample the RVs without parent – i.e., the data bits $b_0^{[m]}, \dots, b_3^{[m]}$ – then sample the check bits, and lastly sample the received bits according to the conditional distribution $p(r_i|b_i)$. Gather the data points in an appropriate data structure, $\mathcal{D} = (\mathbf{d}^{[1]}, \dots, \mathbf{d}^{[M]})$.

¹Of course, if check bit factor parameters are not either 0 or 1, then we have a very strange encoding scheme, since check bits will be generated probabilistically and they might not tell us much about the values of the data bits. However, for this question we are going to pretend that these strange encoding schemes are possible so that we can perform MLE for discrete RVs in the same way as Koller discusses in her videos.

- (c) (On paper) Calculate the likelihood function, $p(\mathcal{D}|\Theta)$, decompose it into a number of local likelihood functions, and write each in terms of the unknown parameters and the number of each type of observations. For example, the local likelihood function for the received bits, $\prod_{m=1}^M \prod_{i=0}^6 p(r_i|b_i : \theta_e)$, should only be written in terms of θ_e , $\#(r_i \neq b_i)$ and $\#(r_i = b_i)$.
- (d) (On paper) Find the parameters that maximise the likelihood (in terms of the number of each type of observation) by solving for $\frac{\partial}{\partial \theta_k} \log p(\mathcal{D}|\Theta) = 0$ for each parameter $\theta_k \in \Theta$.
- (e) (In code) Now process your sampled data points to calculate the ML parameter estimates, Θ^{ML} . Compare the estimated parameters with the actual parameters.
- (f) (In code) Repeat the sampling of data points and parameter estimation, but vary the number of data points. What effect does the number of data points have on the accuracy of the parameter estimates? Under which conditions is it not possible to estimate certain parameters, and why does this happen?

Answers:

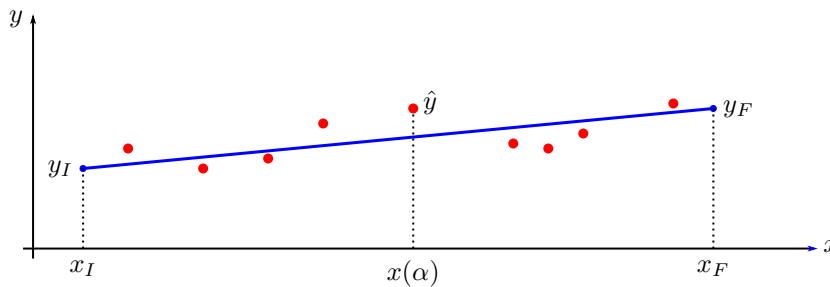
(d)

$$\begin{aligned}\theta_e^{\text{ML}} &= \frac{\sum_{i=0}^6 \#(r_i \neq b_i)}{\sum_{i=0}^6 \#(r_i = b_i) + \sum_{i=0}^6 \#(r_i \neq b_i)} = \frac{\sum_{i=0}^6 \#(r_i \neq b_i)}{7M} \\ \theta_{b_4=1|b_0=0,b_1=0,b_2=0}^{\text{ML}} &= \frac{\#(b_4 = 1, b_0 = 0, b_1 = 0, b_2 = 0)}{\#(b_0 = 0, b_1 = 0, b_2 = 0)} \\ \theta_{b_4=1|b_0=0,b_1=0,b_2=1}^{\text{ML}} &= \frac{\#(b_4 = 1, b_0 = 0, b_1 = 0, b_2 = 1)}{\#(b_0 = 0, b_1 = 0, b_2 = 1)} \\ &\vdots\end{aligned}$$

2 MLE for a simple regression example

In this question, we'll use the Gaussian concepts of Week 7 to fit a line to a number of data points using maximum likelihood estimation. We'll also revisit this problem in Week 9 and Week 10.

Consider the following setup, where a number of noisy measurements (red) are taken of points on a line (blue).



The two endpoints of the line are given by the coordinates (x_I, y_I) and (x_F, y_F) . The x -coordinate of a measurement, x , is known, and it is parameterised by $\alpha \in [0, 1]$ as follows:

$$x = (1 - \alpha)x_I + \alpha x_F. \quad (1)$$

A measurement $\hat{y}$ is modelled as the y -coordinate of the line at x plus zero-mean Gaussian noise, or

$$\hat{y} = (1 - \alpha)y_I + \alpha y_F + v, \quad (2)$$

where $v \sim \mathcal{N}(0, \sigma_v^2)$. The x -coordinates of the endpoints, x_I and x_F , are known. However, the y -coordinates of the endpoints, y_I and y_F , are unknown and are the parameters we are trying to estimate; that is,

$$\Theta = (y_I, y_F). \quad (3)$$

There are M measurements, and the data is therefore given by

$$\mathcal{D} = (\hat{y}^{[1]}, \dots, \hat{y}^{[M]}). \quad (4)$$

We assume that for each $\hat{y}^{[m]}$, its corresponding x parameterisation $\alpha^{[m]}$ is known. The objective of this question is to find the most likely line, Θ , given the measurements, $\mathcal{D}$. Since you again do not need `emdW`'s functionality for this question, you do not need to use C++. [To help you check your work, the answer at an intermediate point is given at the end of this question in blue.](#)

- (On paper) Draw the BN for this model with M measurements using plate notation.
- (On paper) Write $p(\hat{y}^{[m]}|y_I, y_F)$ as an exponential function i.t.o. the data point (or observed random variable) $\hat{y}^{[m]}$, the unknown parameters y_I and y_F , and the known parameters $\alpha^{[m]}$ and σ_v^2 . Use it to construct $p(\mathcal{D}|\Theta)$. Now set up $\log p(\mathcal{D}|\Theta)$ and manipulate it until it is written as a sum of quadratic terms i.t.o. $\hat{y}^{[m]}$, y_I and y_F (and which also includes the known parameters σ_v^2 and $\alpha^{[m]}$).
- (On paper) Derive $\Theta^{\text{ML}} = \arg \max_{\Theta} \log p(\mathcal{D}|\Theta)$ by using the following procedure: Calculate $\frac{\partial}{\partial y_I} \log p(\mathcal{D}|\Theta) = 0$ and manipulate it until you can write it in the form

$$\mathbf{a}_1^T \begin{bmatrix} y_I \\ y_F \end{bmatrix} = b_1, \quad (5)$$

where $\mathbf{a}_1$ and b_1 are written in terms of known quantities. Similarly, calculate $\frac{\partial}{\partial y_F} \log p(\mathcal{D}|\Theta) = 0$ and write it as

$$\mathbf{a}_2^T \begin{bmatrix} y_I \\ y_F \end{bmatrix} = b_2. \quad (6)$$

Now combine Equations 5 and 6 into a single matrix equation and solve for Θ^{ML} .

- (In code) Generate a number of measurements by first uniformly sampling $\alpha^{[m]}$ from the interval $[0, 1]$ and then sampling $\hat{y}^{[m]}$ according to Equation 2 (choose an appropriate measurement noise variance σ_v^2). Plot both the measurements and the actual line.
- (In code) Calculate Θ^{ML} for the data generated in (d) and plot the estimated line on the same plot as the measurements and actual line.
- (In code) Repeat the sampling of measurements and subsequent parameter estimation, but vary the number of measurements. What effect does the number of measurements have on the accuracy of the line fit? What is the lowest number of measurements you can use to fit a line?
- (In code) Repeat the sampling of measurements and subsequent parameter estimation, but vary the measurement noise covariance σ_v^2 while keeping the number of measurements fixed. Does the measurement noise influence the parameter estimation calculations? How does the measurement noise influence the accuracy of the line fit?

Answers:

(c)

$$\begin{aligned} & \left(\sum_{m=1}^M \begin{bmatrix} (1 - \alpha^{[m]})^2 & \alpha^{[m]}(1 - \alpha^{[m]}) \\ \alpha^{[m]}(1 - \alpha^{[m]}) & (\alpha^{[m]})^2 \end{bmatrix} \right) \begin{bmatrix} y_I^{\text{ML}} \\ y_F^{\text{ML}} \end{bmatrix} = \sum_{m=1}^M \begin{bmatrix} (1 - \alpha^{[m]})\hat{y}^{[m]} \\ \alpha^{[m]}\hat{y}^{[m]} \end{bmatrix} \\ \Rightarrow & \begin{bmatrix} y_I^{\text{ML}} \\ y_F^{\text{ML}} \end{bmatrix} = \left(\sum_{m=1}^M \begin{bmatrix} (1 - \alpha^{[m]})^2 & \alpha^{[m]}(1 - \alpha^{[m]}) \\ \alpha^{[m]}(1 - \alpha^{[m]}) & (\alpha^{[m]})^2 \end{bmatrix} \right)^{-1} \left(\sum_{m=1}^M \begin{bmatrix} (1 - \alpha^{[m]})\hat{y}^{[m]} \\ \alpha^{[m]}\hat{y}^{[m]} \end{bmatrix} \right) \end{aligned}$$

3 Assessment: Class test

There will be short, written class test at the end of the practical period in the practical venue (see STEMLearn for details). If you have worked through the practical assignment and understood everything, then the test should be easy!